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lpjmlkit: A toolkit for operating LPJmL and model-specific data processing

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Summary

The *lpjmlkit* R package (Breier et al., 2023) is an open source software that is developed for handling the open source dynamic global vegetation model (DGVM) LPJmL. It contains two main modules. One, *LPJmL Runner*, provides the functionality to create multiple model configurations and start the corresponding simulations either on a personal computer or on an HPC (High Perfomance Computing) cluster with SLURM (Simple Linux Utility for Resource Management) support, in both cases requiring a working LPJmL installation. The other, *LPJmL Data*, offers a generic function that supports reading both simulation output and model input data in multiple file formats used by LPJmL. The associated data class LPJmLData contains both the data and the corresponding metadata to ensure data integrity within a single instance. LPJmLData objects act as data containers that provide modification functions such as subsetting or transformations of the data. LPJmLData objects can be exported into various other common R data formats. In addition to these modules, other functions are included to facilitate common use cases of LPJmL. This article introduces *lpjmlkit*, an R package that serves as an interface to LPJmL to simplify direct work with the model and to enable new generic software developments based on LPJmL simulations or data.

Statement of need

A simple interface facilitates the use of software and improves accessibility, user experience, and overall adoption. Scientific software should also be findable, accessible, interoperable and reusable, according to the FAIR principles for research software (Barker et al., 2022). Numerical models based on process-based approaches and implemented in low-level programming languages are often characterised by a long history and long development cycles, that affect operability. These models therefore often lack simple and reliable interfaces that are also FAIR (Barker et

al., 2022; Wilson et al., 2014). LPJmL is a well-established dynamic global vegetation model (DGVM), widely used in the scientific community. A DGVM simulates potential vegetation shifts and associated biogeochemical and hydrological cycles in response to changes in climate. (Cramer et al., 2001) DGVMs are typically used to simulate the effects of future climate change on natural vegetation and its carbon and water cycles. In this context LPJmL has been used for more than a decade and was employed by researchers to conduct numerous studies in various research areas related to the terrestrial biosphere. To this end, the original DGVM LPJ (Sitch et al., 2003) was extended by adding an improved representation of the hydrological cycle (Gerten et al., 2004), by implementing managed land components, forming "LPJmL" (LPJ with *m*anaged *L*and)) (Bondeau et al., 2007; Lutz et al., 2019; Rolinski et al., 2018; Schaphoff et al., 2018), and by including the nitrogen cycle (Bloh et al., 2018). This facilitated broader, interdisciplinary studies such as the work of Gerten et al. (2020), which answered the question of whether it is possible to feed ten billion people within four planetary boundaries, or studies that implemented features of sustainable agriculture in LPJmL (Herzfeld et al., 2021; Porwollik et al., 2022) that had not been simulated before within a DGVM. There are many other examples of different scientific studies based on LPJmL simulations, all using individual scripts to create inputs for the model or analyze outputs.

Unlike its sister model LPJ-GUESS (Bagnara et al., 2019) or other models, such as MAgPIE (Dietrich et al., 2019), LPJmL was never equipped with standardised interfaces for higher level programming languages to either run simulations or read and process input or output data. The lack of a standardized interface means that both beginners and experienced users need to constantly develop their own custom scripts and tools for what should be routine (data) processing steps. Tools shared informally between individual users often have limited scope and as such limited re-usability. They lack documentation and are often not well tested or maintained, nor released to a wide user base in a transparent manner.

lpjmlkit was developed to address these problems and at the same time create a standard in the handling of LPJmL to improve the management of simulation experiments and allow for better documentation and reproducibility of studies with LPJmL. By using the *LPJmL Runner* functionality, model configurations are stored in a single and unique configuration file that references the exact model version of LPJmL used for the simulations to achieve reproducible results. *LPJmL Data* subsequently ensures a generic standard for version-independent processing of LPJmL (output) data. This way, *lpjmlkit* serves as a user and programming interface to LPJmL and provides an easy-to-use basis for interaction with LPJmL in a simple R script as well as for further software development based on LPJmL, for example model calibration, benchmarking or indicator development.

Package features

lpjmlkit is an R package that contains two main modules, each of which contains supplementary documentation in form of a guide in addition to the package documentation:

- *LPJmL Runner* vignette
- *LPJmL Data* vignette

Together, these modules can be used to configure and run LPJmL model simulations as well as read and process the resulting data.

LPJmL Runner

The Runner module is designed to operate LPJmL on Unix-based operating systems that have a working LPJmL installation (version 4 or higher) and includes four key functions.

The basic idea is to create tables for the parameters and settings to be changed from a default configuration file for single runs or any number of related simulations, e.g. multi-scenario or uncertainty/sensitivity analyses. The creation of these tables is based on the concept of `tidy`

data with columns as parameters/settings (variables) and rows as simulations (observations) (Wickham, 2014). In order to facitlitate the execution of these simulations, the function `write_config()` handles the formatting of the corresponding `tidy data` tables into LPJmL compatible JSON configuration files.

LPJmL simulations can either be run sequentially on single CPUs with `run_lpjml()`, which is particularly suitable for testing the model or for focusing on small regions. In addition, simulations can be submitted to an HPC cluster equipped with the SLURM workload manager using `submit_lpjml()`, which is particularly useful for computing multiple global simulations in parallel mode using multiple CPUs.

LPJmL Data

While the module *LPJmL Runner* covers the whole range of running LPJmL simulations, `LPJmL Data` provides the tools for the subsequent data analysis part. LPJmL simulations output data in a raw binary data format designed for memory efficient data structures. The LPJmL model can write metadata for each output file, either as a file header or as an additional metadata file in JSON format. Using `read_io()`, LPJmL raw output data can be read into a standardised data format called `LPJmLData`, which makes use of the corresponding metadata to correctly represent the data arrays (dimensions, units, etc.).

This data format is designed to faciliitate the functionalities commonly used by LPJmL users for data analysis. In addition to the descriptive statistics that an LPJmLData object displays by default, there is a `plot()` method to easily display the the data for visual inspection. `LPJmLData` objects can be transformed into various common spatial and temporal formats with the funciton `transform()`. For example, the one-dimensional cell vector can be transformed into a longitude-latitude matrix to allow further spatial operations such as "nearest neighbour" subsetting, using coordinate pairs. The temporal dimension can be decomposed into further sub-time dimensions, which is particularly useful for high temporal resolutions such as monthly or daily data covering several years. Several export functions allow for easy conversion of the LPJmLData format to other common data formats such as `SpatRaster` (`as_terra()`), `tibble` (`as_tibble()`) or simply `array` (`as_array()`). Here, in addition to subsetting, the data can also be aggregated using common functions such as `mean()` or `sum()`.

Especially in combination, these different functions can greatly increases the utility for data analysis. For example, a combination of the transformation of the read-in discharge into sub-time dimensions and latitude-longitude matrix and the subsequent subsetting of the summer months of the northern hemisphere and the aggregation over years and months allows the following presentation.

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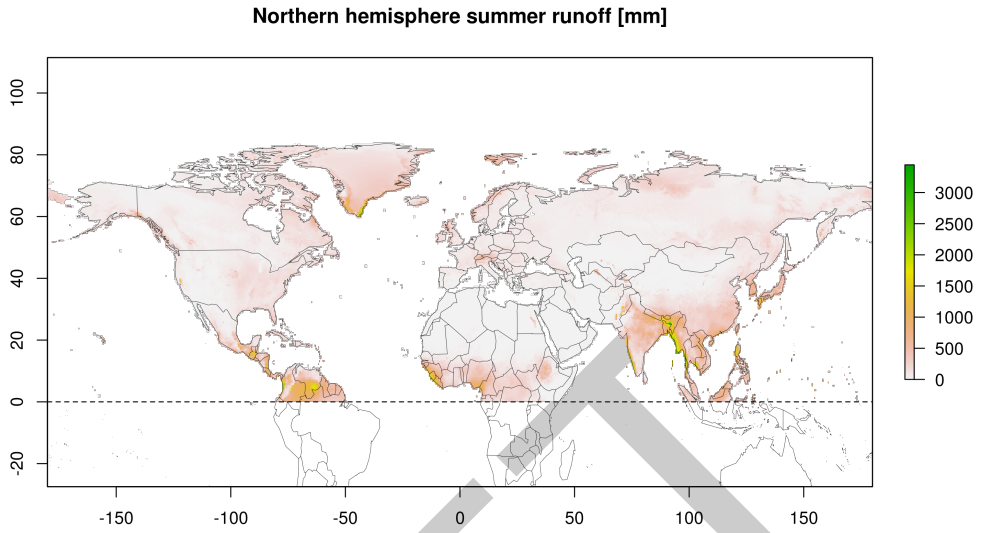

Figure 1: Exemplary LPJmLData plot of variable runoff generated by functions read_io, transform, subset and plot

Miscellaneos

The *lpjmlkit* package also includes additional functions to support different applications that involve handling LPJmL or related data.

For example, the function `calc_cellarea()` calculates the area of the grid cells in an `LPJmLData` object or for any vector of latitude coordinates representing a regular grid with a set spatial resolution.

While `read_io()` is the main function to access LPJmL input and output files, *lpjmlkit* also includes a number of functions providing low-level access to the binary LPJmL file format: `read_header()`, `create_header()`, and `write_header()` allow to retrieve/generate/save file headers, while `get_headersize()` and `get_datatype()` return information about the header and about the data type used in different LPJmL files, respectively. These low-level access functions can form the basis for new applications, for example, to generate new LPJmL input files.

Documentation & License

lpjmlkit is an open-source software package for operating LPJmL and processing related data that provides comprehensive documentation and vignettes. The documentation is available online at https://pik-piam.r-universe.dev/lpjmlkit and includes instructions for installation, usage, and examples. The package is licensed under the GNU Affero General Public License (AGPL-3.0), which grants users the freedom to access, use, and modify the code, and ensures that any modifications or derivative works are also available under the same license, allowing for continued collaboration and development within the community. The source code is available on GitHub at https://github.com/PIK-LPJmL/lpjmlkit, where users can also report issues and suggest improvements.

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